ADEQUACY OF AN ADEQUATE MODULE IS LOCAL FOR THE TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Adequacy is preserved by base change of the test algebra. Same chapter, Lemma lemma-base-change-adequate, invoked at proof lines 1722-1723. Makes the functors over the refined cover adequate.
- Adequate functors over a finite product of rings correspond to tuples of adequate functors. Same chapter, Lemma lemma-adequate-product, invoked at proof lines 1724-1726. Assembles the refined functors into one functor over the product ring A'.
- Adequacy descends along a tau-covering of affine schemes. Same chapter, Lemma lemma-adequate-descent, invoked at proof lines 1732-1734. Descends adequacy from A' to A.

Source reference. Original lemma 06VH: sources/stacks-adequate/pinned-original-source.tex, source lines 1003–1003. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06VB: sources/stacks-adequate/pinned-original-source.tex, source lines 989–989. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06V9: sources/stacks-adequate/pinned-original-source.tex, source lines 953–953. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{F}$ be an adequate $\mathcal{O}$-module on $(Sch/S)_\tau$. For any affine scheme* $\mathrm{Spec}(A)$ *over S the functor $F_{\mathcal{F},A}$ is adequate.*

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Proof. Let $\{\mathrm{Spec}(A_i) \to S\}_{i \in I}$ be a τ-covering such that $F_{\mathcal{F},A_i}$ is adequate for all $i \in I$. We can find a standard affine τ-covering $\{\mathrm{Spec}(A'_j) \to \mathrm{Spec}(A)\}_{j=1,\ldots,m}$ such that $\mathrm{Spec}(A'_j) \to \mathrm{Spec}(A) \to S$ factors through $\mathrm{Spec}(A_{i(j)})$ for some $i(j) \in I$. Then we see that $F_{\mathcal{F},A'_j}$ is the restriction of $F_{\mathcal{F},A_{i(j)}}$ to the category of A'_j-algebras. Hence $F_{\mathcal{F},A'_j}$ is adequate by Lemma 06V9. By Lemma 06VB the sequence $F_{\mathcal{F},A'_j}$ corresponds to an adequate "product" functor F' over $A' = A'_1 \times \ldots \times A'_m$. As $\mathcal{F}$ is a sheaf (for the Zariski topology) this product functor F' is equal to $F_{\mathcal{F},A'}$, i.e., is the restriction of F to A'-algebras. Finally, $\{\mathrm{Spec}(A') \to \mathrm{Spec}(A)\}$ is a τ-covering. It follows from Lemma 06VH that $F_{\mathcal{F},A}$ is adequate. $\square$